

Deliverable 1

Project 2 Outline

Overview

Your first Project 2 deliverable is due **Tuesday, March 17 at 10:00 PM**. It should be submitted as a concise **2–3 page PDF file** outlining your project, written in 12-point font and single-spaced. This outline should include the following sections:

1. Project Title

- Provide a concise, descriptive, and informative title for your project.

2. Group Members

- List the names and student IDs of all group members.

3. Background and Motivation

- State why your group finds the two chosen violation types interesting or practically relevant.

4. Chosen Simulation Studies

- Identify which two studies (A–D) your group has selected for Part I and justify your choices.
- For each chosen study, briefly describe the violation it investigates and your plan to simulate.

5. Research Questions and Objectives

- State the two core questions the project investigates:
 1. How badly do classical 95% CIs and p-values for β_1 fail under your chosen violations?
 2. Does the bootstrap provide a reliable remedy?

- List any additional group-specific hypotheses (e.g., whether larger n restores coverage, or whether bootstrap p-values outperform classical ones under a specific violation).

6. Data Source and Description (Part II)

- Name the dataset your group has chosen for the real data analysis and provide its source (with URL or R package).
- Identify the response variable (Y) and the primary predictor (X), and give a brief justification for this choice.
- Provide a short summary of the dataset: number of observations, relevant variable ranges, and any known data limitations or preprocessing needed.
- Note which assumption violations you expect to find, and how they connect to your chosen simulation studies.

7. Timeline and Roles

- Provide a rough schedule for each major milestone.
- Assign clear tasks and responsibilities to group members.
- **Example Milestones:**
 1. Project Outline Submission (Deliverable 1): Due **Tuesday Mar 17 at 10:00 PM**
 2. Part I Simulation: error generation, coverage table, histograms, and p-value comparison — **Lab, Wednesday March 18**
 3. Part II Real Data Analysis: EDA, OLS, bootstrap CI and p-value, diagnostics — **Lab, Wednesday March 25**
 4. Progress Report — Draft slides + R code + dataset (Deliverable 2): Due **Monday March 30 at 10:00 PM**
 5. Part III Reflection and report finalisation — **Lab, Wednesday April 1**
 6. Final Written Report (PDF) + R script + dataset (Deliverable 3): Due **Tuesday April 7 at 10:00 PM**

7. Resubmission deadline (groups receiving R or N): **Wednesday April 29**

- **Example Tasks or Responsibilities:**

- 4 group members:
 1. Introduction + dataset selection and EDA (Part II)
 2. Part I simulation — Study 1 (scenarios, coverage table, histograms)
 3. Part I simulation — Study 2 (scenarios, coverage table, bootstrap p-values)
 4. Diagnostics, Part III Reflection, Conclusion + report editing
- 3 group members:
 1. Introduction + Part I simulation (first studies)
 2. Part II real data analysis (EDA, OLS, bootstrap CI and p-value, diagnostics)
 3. Part I simulation (second studies) + Conclusion + report editing